

MERRILL KEATING

Seattle, WA • 206.335.4832 • merrill@merrillkeating.com
linkedin.com/in/merrill-keating • github.com/mcheyenne7-coder

SUMMARY

Mechanical engineer with 6+ years of hands-on experience in electromechanical design, prototyping, and system integration. Background spans NASA research, competition robotics, and university rover design — all emphasizing design-build-test cycles with real hardware. Trained in biomechanics, ergonomics, and prosthetics design. Strong in FEA, GD&T, mechanism design, and embedded systems. Published author on AI ethics and human-machine collaboration.

TECHNICAL SKILLS

CAD & Analysis: SolidWorks, AutoCAD, Inventor, FEA (structural, thermal, modal, fatigue), GD&T (ASME Y14.5), tolerance analysis

Electromechanical: Stepper motors, brushless DC motors, servos, actuators, solenoids, power systems, battery management, motor testing & selection, current limiting

Embedded Systems: Microcontrollers, Arduino, Raspberry Pi, ROS, sensor integration, UART, I2C, Bluetooth, WiFi

Test & Instrumentation: Test fixture design, test planning & execution, IMU data acquisition, signal processing, data analysis

Fabrication: 3D printing, CNC machining, mill, lathe, bandsaw, grinder, CO2 laser cutting, soldering, wiring harnesses

Programming: Python, MATLAB, C++, JavaScript, ROS

ENGINEERING EXPERIENCE

NASA Space Grant Research — AIRLAB | Research Fellow | University of Washington | 2021–2023

- Designed self-resettable hopping robot mechanisms for extraterrestrial terrain traversal
- Tested brushless motors with current limiting to select optimal actuators; evaluated power and battery requirements
- Performed hands-on soldering and circuit assembly for robot prototypes
- Developed Python simulation modeling Mars entry, deceleration, and aerobraking trajectories
- Performed FEA simulations for structural loads during reentry and shock absorption
- Presented findings at UW 25th Annual Undergraduate Research Symposium

Wave Energy Conversion Research (Capstone) | Lead Engineer | UW + National Lab | 2023–2024

- Led design, fabrication, and testing of spring-integrated pendulum system for wave energy conversion
- Built electromechanical test rig with microcontroller-driven actuators, embedded sensors, and power interface
- Applied GD&T and tolerance analysis in 3D CAD to ensure robustness under dynamic loads
- Integrated ROS for data acquisition; developed Python code for IMU signal processing
- Collaborated with national lab researchers on sensor selection, test planning, and system integration

UW Husky Robotics — Chassis Subsystem | Team Member | 2020–2021

- Designed chassis and wheelbase for University Rover Challenge Mars-terrain simulation
- Worked with stepper motors, brushless DC motors, and servos for drive and articulation systems
- Integrated ROS for control systems; contributed to sensor communication architecture
- Conducted materials research for optimal wheel composition under simulated Martian conditions

FIRST Robotics Team 4915 | Mechanical Lead | 2018–2020

- Designed and built competition robots with drive trains, mechanisms, servos, and microcontroller-based control
- Hands-on fabrication: CNC machining, mill, lathe, laser cutting, 3D printing, soldering, wiring harnesses
- Applied GD&T and tolerance analysis in CAD for precision mechanism design
- Awards: District Event Winner, Engineering Inspiration Award (2x), Industrial Design Award

West Sound CoderDojo / WSTA | Lead Mentor & Program Developer | 2015–Present

- Developed and taught embedded systems curriculum using Arduino, Raspberry Pi, and space exploration kits
- 10 years mentoring learners ages 7-70 in programming, robotics, and hardware fundamentals

Keating Consulting Services, Inc. | IT Consultant | 2019–Present

- Device deployment, network configuration (WiFi, Bluetooth), and system troubleshooting for K-12 districts
- Client-facing project management; database development and system migrations

EDUCATION

University of Washington — B.S. Mechanical Engineering | August 2025

Early entrance program (began at 15) • Resident Adviser 2021–2023

Relevant Coursework: Biomechanics, Prosthetics Design, Human Ergonomics, FEA, Structures, Machine Design, Microcontroller Systems (ME 478), Machining

ADDITIONAL

Human-Machine Interaction: Published 5 books on AI consciousness, ethics, and collaboration. Strong interest in VR and human-robot ergonomics.

Leadership: Founded The Power of 100 Girls & Girls Ignited (226K+ community). Board Member, West Sound Technology Association. Organized TEDx events.

Recognition: Grace Hopper Award (2025) • USA Today Women of the Year, WA (2024) • Outstanding Young Philanthropist (2024)